INTELLILINK GATEWAY™

Field Case Study

Governance Restoration for Enterprise Satellite Connectivity

Document Control

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- **Data retention:** No enterprise customer data retained beyond transient packet forwarding

Architect Attribution

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Case Study Overviw

This Field Case Study documents a controlled deployment of the Intellilink Gateway™ control plane architecture in a real operational environment. The objective of the validation exercise was to determine whether satellite-connected enterprise traffic could be re-anchored within a domestic Internet Service Provider (ISP) governance domain while maintaining the resilience and accessibility benefits of low-Earth-orbit (LEO) satellite connectivity.

The deployment focused on three primary validation objectives:

- Routing Control
- Governance anchoring
- Tunnel Stability

The exercise was conducted as a single-day validation window, using live enterprise traffic while applying strict anonymization to participating entities.

Test Environment Description

The validation deployment consisted of the following components:

- **Enterprise Site:** An operational enterprise network generating normal Internet traffic.
- **Satellite Access Layer:** Connectivity provided by a low-Earth-orbit satellite Internet service (Starlink).
- **Gateway Node:** An Intellilink Gateway Agent Node deployed adjacent to the enterprise satellite terminal.
- **Domestic ISP Point-of-Presence (PoP):** An ISP PoP serving as the governance anchor point where traffic decapsulation, policy enforcement, and routing occured.
- **External Network:** Standard Internet destinations accessed through the ISP's external transit connectivity.

Anonymised Notice

All infrastructure participants were anonymized for this public document. No enterprise customer data was retained beyond transient packet forwarding.

Deployment Architecture

The deployment architecture introduced an overlay governance control plane on top of an existing satellite transport layer.

An enterprise traffic originating within the enterprise LAN was first processed by the Intellilink Gateway Agent Node, which established an encrypted tunnel toward the Intellilink Compliance Gateway PoP (ICGPoP) located within the domestic ISP domain.

Satellite Connectivity served purely as transport underlay, while governance functions including traffic decapsulation, policy enforcement, logging, and regulatory anchoring occured within the ISP PoP environment.

This architecture effectively seperated connectivity transport from governance enforcement.

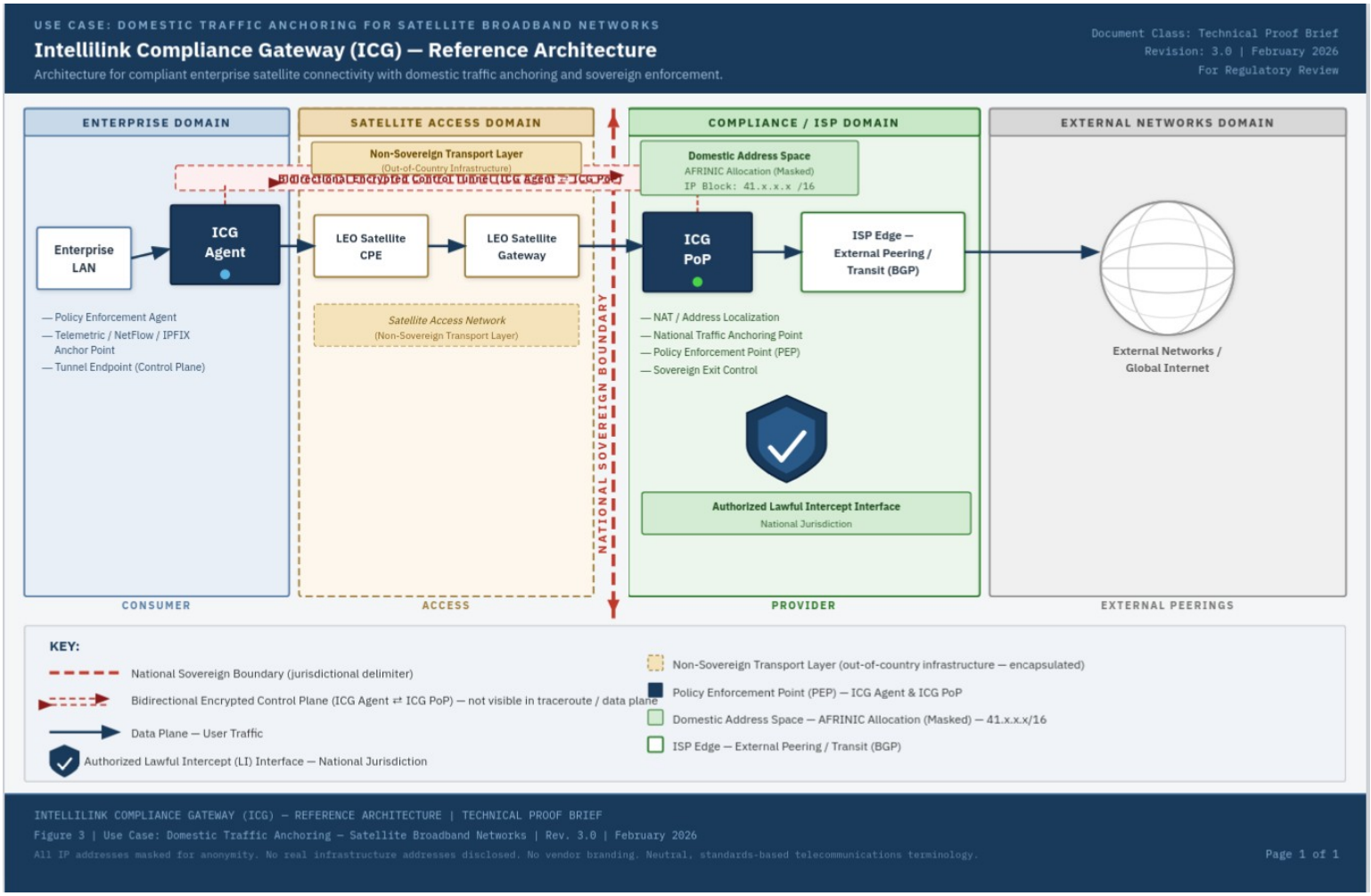

Figure 1: Intellilink Gateway™ Architecture Used During Field Validation

This diagram illustrates the logical architecture deployed during the validation exercise, showing the separation between the satellite transport layer and the governance control plane anchored within the ISP domain.

Routing Behaviour Before Deployment

Prior to deploying the Intellilink Gateway™ control plane, enterprise traffic exited the satellite network directly through upstream satellite provider routing paths.

In this configuration, traffic did not traverse a domestic ISP governance anchor point prior to reaching the broader Internet. This behavior reduced visibility from a domestic network governance perspective because:

- Upstream accountability was ambiguous
- Traffic inspection points were absent within national networks
- Routing behavior was externally determined by satellite provider infrastructure

Traceroute measurements taken during this stage confirmed that traffic exited the satellite transport network before entering external transit providers.

USE CASE: DOMESTIC TRAFFIC ANCHORING FOR SATELLITE BROADBAND NETWORKS

Intellilink Compliance Gateway (ICG)
Traffic Anchoring — Before / After Routing Comparison

Figure 2 — Traceroute-Based Routing Validation | Source: Enterprise LAN (192.168.x.x, Lusaka) → Destination: speedtest.mtn.zm (41.223.117.81)

For Regulatory Review

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Routing Behaviour Comparison Demonstrates how ICG deployment anchors enterprise satellite traffic within national legal jurisdiction, restoring domestic policy enforcement and lawful intercept capability.

⚠ BEFORE ICG
17 hops / 242 ms
8 foreign ASN hops | No LI capability

✔ AFTER ICG
7 hops / 302 ms
0 foreign ASN hops | Full LI capability

Note on latency delta (+60 ms): *Increased latency reflects the overhead of domestic traffic anchoring, NAT processing, and compliance enforcement at the ICG PoP. This is consistent with the security and regulatory objectives of the deployment.*

LEGEND Data Plane (User Traffic) — — — Encapsulated Tunnel (Data Plane) — — — Control Plane (Encrypted — not traceroute-visible) Foreign / Non-Sovereign ASN Domestic ASN (Zambia)

Encapsulated Hop (not data-plane visible) [ ] Sovereign Network Boundary

⚠ BEFORE ICG DEPLOYMENT — NON-SOVEREIGN ROUTING (NON-COMPLIANT)

DATA PLANE ORIGIN
ENTERPRISE LAN
192.168.1.79 — LUSAKA, ZAMBIA

ENTERPRISE DOMAIN

No Compliance Gateway
Traffic exits unencapsulated

SATELLITE ACCESS DOMAIN — NON-SOVEREIGN TRANSPORT

LEO Satellite CPE

Satellite Provider Network

LEO Satellite Gateway
Foreign egress — jurisdiction exit

⚠ SOVEREIGN NETWORK BOUNDARY — TRAVERSED WITHOUT CONTROL

EXTERNAL NETWORKS — INTERNATIONAL TRANSIT (FOREIGN ASNS)

Foreign ASN Transit
206.224.73.x / 154.66.247.x / 149.6.x.x

TRACEROUTE — BEFORE ICG | ENTERPRISE LAN → 41.223.117.81

1	192.168.1.1	112.817 ms	LAN Default GW
2	100.64.0.1	171.163 ms	CGNAT (Sat. CPE)
3	172.16.250.86	170.346 ms	Satellite Provider PoP
⚠ Sovereign Network Boundary crossed — foreign ASN transit begins			
4	206.224.73.68	170.073 ms	Foreign ASN
5	206.224.73.52	170.196 ms	Foreign ASN
6	102.68.115.226	170.877 ms	Foreign ASN
7	154.66.247.66	204.070 ms	Foreign transit
8	154.66.247.106	62.200 ms	Foreign transit
9	154.66.247.99	203.870 ms	Foreign transit
10	* * *		No response (filtered)
11	149.6.144.125	216.014 ms	Foreign transit
↩ Re-entry into Zambia domestic ASN			
12	41.181.105.163	215.952 ms	Carrier Edge
13	41.181.244.162	339.148 ms	ISP Core
14	41.181.244.205	201.123 ms	ISP Core

✔ AFTER ICG DEPLOYMENT — DOMESTICALLY ANCHORED ROUTING (COMPLIANT)

DATA PLANE ORIGIN
ENTERPRISE LAN
192.168.101.5 — LUSAKA, ZAMBIA

ENTERPRISE DOMAIN

ICG Agent (Compliance Gateway)
PEP TUNNEL ENDPOINT
Policy enforcement · Telemetry · Routing control

Control Plane (Encrypted) — ICG Agent ⇄ ICG PoP
Tunnel management · Policy updates · Telemetry
Not visible in traceroute output

SATELLITE ACCESS DOMAIN — NON-SOVEREIGN TRANSPORT (TUNNEL-ENCAPSULATED)

LEO Satellite CPE

Encapsulated tunnel (data plane not exposed)

Satellite Provider Network

LEO Satellite Gateway
Transit only — payload opaque to provider

TRACEROUTE — AFTER ICG | ENTERPRISE LAN → 41.223.117.81

1	192.168.101.1	2.020 ms	ICG Agent — LAN GW
→ Satellite segment (encapsulated tunnel — hops not data-plane visible)			
2	172.16.10.1	246.031 ms	ICG PoP ingress — tunnel decap
3	41.x.x.x	243.932 ms	NAT egress — Local ISP ASN (masked)
4	* * *		No response (ICMP filtered)
→ Local IXP / MTN domestic transit			
5	192.32.209.224	291.902 ms	Local IXP
6	41.223.119.202	269.292 ms	MTN Edge
7	41.223.117.81	302.121 ms	DESTINATION ✓

■ Satellite transit hops are encapsulated within the ICG tunnel and are not visible as individual traceroute hops. This is expected behaviour and does not indicate routing anomalies.

No lawful intercept · No domestic policy enforcement

↩ RE-ENTRY INTO NATIONAL JURISDICTION

DOMESTIC ISP DOMAIN (RE-ENTRY — ZAMBIA ASN)

Carrier / ISP Edge

ISP Edge

DATA PLANE DESTINATION
SPEEDTEST.MTN.ZM
41.223.117.81 — MTN ZAMBIA

15	41.181.190.133	193.452 ms	ISP Distribution
16	41.223.119.202	238.447 ms	MTN Edge
17	41.223.117.81	242.281 ms	DESTINATION ✓

COMPLIANCE ASSESSMENT — NON-COMPLIANT

- ✗ 8 hops through foreign ASN infrastructure (hops 4–11); traffic exits national legal jurisdiction.
- ✗ No domestic address localization — source IP is not anchored to a national ASN at egress.
- ✗ No lawful intercept capability during foreign ASN transit segment.
- ✗ No policy enforcement point present in path before international egress.
- ✗ Traffic re-enters domestic jurisdiction only at hop 12, without any compliance validation.

✔ SOVEREIGN NETWORK BOUNDARY — TRAFFIC ANCHORED DOMESTICALLY

COMPLIANCE / ISP DOMAIN — NATIONAL POLICY ENFORCEMENT ZONE

ICG PoP — Domestic Anchor Point
NAT / ADDRESS LOCALIZATION PEP
LAWFUL INTERCEPT
National ASN egress · Zambia legal jurisdiction · AFRINIC address space

ISP Edge — National ASN (Lusaka)

Local IXP (Domestic Peering)

MTN Edge

DATA PLANE DESTINATION
SPEEDTEST.MTN.ZM
41.223.117.81 — MTN ZAMBIA

COMPLIANCE ASSESSMENT — COMPLIANT

- ✔ Zero foreign ASN hops observed in traceroute. All intermediate addresses within domestic ASN space.
- ✔ Address localization enforced at ICG PoP via NAT; public egress uses domestic ASN address space.
- ✔ Lawful intercept capability present at ICG PoP, within national legal jurisdiction.
- ✔ Policy Enforcement Point (PEP) active at both ICG Agent (enterprise) and ICG PoP (compliance domain).
- ✔ Satellite segment traffic is encapsulated — provider has no access to payload data plane.

POLICY COMPLIANCE OUTCOMES — ICG DEPLOYMENT

- ✔ Domestic traffic anchoring
- ✔ Lawful intercept capability
- ✔ Domestic ASN public egress
- ✔ National jurisdiction enforced
- ✔ Address localization (NAT)
- ✔ Authority visibility restored

LATENCY NOTE — BEFORE VS. AFTER

The observed increase in end-to-end latency (+60 ms; 242 ms → 302 ms) is attributable to the domestic anchoring path via the ICG PoP, including tunnel decapsulation, NAT processing, and local IXP peering. This latency overhead is an inherent and expected consequence of compliance enforcement and is consistent with domestic traffic anchoring obligations.

CONTROL PLANE — TRACEROUTE INVISIBILITY

The encrypted control plane tunnel between the ICG Agent and ICG PoP operates independently of the data plane and is not reflected in traceroute output. The satellite transit segment appears compressed to a single logical hop (hop 1 → hop 2) due to tunnel encapsulation. This is expected and does not indicate missing hops or routing anomalies.

LAWFUL INTERCEPT — REGULATORY NOTE

Lawful intercept (LI) capability is implemented at the ICG PoP, within the national legal jurisdiction. LI interfaces comply with applicable national telecommunications regulations. No interception capability is delegated to the satellite provider or foreign ASN infrastructure.

Figure 2 — Traceroute Routing Validation | Use Case: Domestic Traffic Anchoring — Satellite Broadband

Routing Behaviour After Deployment

Following deployment of the Intellilink Gateway™ Agent Node, enterprise traffic was encapsulated into an encrypted overlay tunnel directed toward the domestic ISP governance PoP.

Upon arrival at the Intellilink Compliance Gateway PoP (ICGPoP), traffic underwent the following processing steps:

- Tunnel termination
- Traffic decapsulation
- Identity binding
- Policy enforcement
- Observability logging

After processing, traffic was routed through the ISP's external peering infrastructure before reaching the public Internet.

Traceroute measurements confirmed that enterprise traffic now entered the domestic ISP routing environment prior to external Internet transit. This demonstrated successful traffic re-anchoring within the domestic governance domain.

Governance Restoration

The deployment demonstrated that enterprise satellite connectivity could be integrated into a conventional ISP governance model without modifying the satellite access infrastructure. The following governance properties were successfully restored:

- **Accountable Upstream Presence:** Enterprise traffic visibly entered a domestic ISP Point of Presence.
- **Address Sovereignty:** Enterprise traffic appeared under ISP-controlled addressing space following tunnel termination.
- **Traffic Observability:**Flow-level inspection and logging were possible within the ISP environment.
- **Policy Enforcement Capability:** Network policy controls could be applied prior to external Internet transit.

These characteristics align with regulatory expectations for accountable Internet traffic flow within national communications frameworks.

ENTERPRISE DOMAIN
(ANONYMIZED)
SATELLITE TRANSPORT DOMAIN
DOMESTIC ISP / COMPLIANCE DOMAIN
(ANONYMIZED)
EXTERNAL NETWORKS
Enterprise LAN / Applications
Enterprise Users
Workstations
Mobile / IoT Devices
Internal Applications
INTELLILINK GATEWAY™
Intellilink Gateway™ — Agent Node
Tunnel Endpoint (Initiator)
Traffic Steering
Policy Enforcement Agent
Tunnel Origination
Enterprise Edge Control
LEO Satellite CPE (Starlink)
Satellite Access Terminal
LEO Transport Interface
LEO Satellite Transport Segment
LEO Satellite Transport Underlay
NATIONAL JURISDICTION BOUNDARY
DOMESTIC ISP POINT OF PRESENCE (ANONYMIZED)
GOVERNANCE ANCHOR
Intellilink Compliance Gateway PoP (ICGPoP)
Tunnel Endpoint (Terminator)
Tunnel Termination
Traffic Decapsulation
Identity Binding
Certificate Validation
Regulatory Anchor Point
Traffic Re-Anchoring
Policy Enforcement Point
Access Control
Traffic Filtering
QoS / Traffic Shaping
Geofencing Controls
Jurisdiction Enforcement
Logging / Observability Platform
NetFlow / IPFIX Export
SIEM Integration
Regulatory Reporting
Immutable Audit Trail
ISP Edge Router / External Peering (ASBR)
BGP Peering
Transit / IX Connectivity
External Networks — Global Internet
Public Routing
TRAFFIC FLOW REFERENCE
DATA PLANE
Enterprise LAN
Intellilink Gateway™ Agent
Starlink CPE
Satellite Transport
ICGPoP
Policy Enforcement Point
ISP Edge / ASBR
Global Internet
ENCRYPTED TUNNEL
Intellilink Gateway™ Agent
ICGPoP (Tunnel Termination)
Bidirectional Encrypted Tunnel (Overlay Control Plane)
SATELLITE UNDERLAY
Starlink CPE
Satellite Transport
ISP PoP
LEO Satellite Transport Underlay
LEGEND
Data Plane Traffic
Encrypted Tunnel (Overlay)
Satellite Underlay
Governance / Policy Enforcement Points
Lawful Intercept Capability Anchor

__Figure 3__ illustrates the anonymized deployment topology used during the validation exercise. Enterprise traffic originates within the enterprise LAN and is steered by the Intellilink Gateway™ Agent through an encrypted overlay tunnel terminating at the Intellilink Compliance Gateway PoP (ICGPoP) within the domestic ISP domain. Upon tunnel termination, traffic undergoes decapsulation, identity binding, and policy enforcement before entering the ISP routing environment. Logging, observability, and governance enforcement occur prior to external Internet transit, restoring regulatory visibility and accountability while preserving satellite transport connectivity.

Operational Observations

Throughout the one-day validation window, the system demonstrated stable operational behavior. Observations included:

- Consistent encrypted tunnel stability
- Predictable routing behavior after governance anchoring
- No observable disruption to enterprise browsing activity
- Successful policy insertion capability at the ISP PoP

Enterprise applications functioned transparently without requiring modifications to end-user systems.

Limitations of the Validation Window

The field validation was intentionally limited in scope. The exercise did not evaluate:

- Commercial-scale enterprise deployments
- Long-duration operational stability
- Lawful intercept system integration
- Enterprise billing models or retail ISP functions

Scope Notice

These areas require additional evaluation beyond the scope of this validation window.

Conclusion

The field validation demonstrates that enterprise satellite connectivity can be integrated into domestic Internet governance frameworks through the use of an overlay control plane architecture. The Intellilink Gateway™ model successfully restored ISP anchoring, routing accountability, and policy enforcement capabilities while maintaining the resilience benefits of satellite connectivity.

These results indicate that satellite innovation and national Internet governance principles can coexist when integration is approached through deliberate architecture design.

Legal & Intellectual Property

- **Legal Disclaimer:** This document describes a technical validation exercise only. It does not constitute regulatory approval, commercial certification, or service guarantees.

  All participating entities have been anonymized.

- **Intellectual Property Notice:** The Intellilink Gateway™ architecture and control plane design are proprietary intellectual property of Intellilink Media LLC™.

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